

Computer Science and Engineering Department, IIIT Kota
Mid Term Examinations, September 2024

Subject: Statistics for Data Science
Duration: 1.5 Hours

Subject Code: CST501
Full Marks: 30

Answer all three questions. Figures at the right margin indicate marks.

1. (a) Write the importance of Gaussian distribution in data science and also define the PDF for multivariate Gaussian distribution. [2]
- (b) What evaluation metrics would you work to gauge the effectiveness of a regression model? [2]
- (c) Which of the following statements are correct? [2]
- (i) Underfitting leads to high training error and high testing error. (ii) Overfitting leads to low training error and low testing error. (iii) High generalization error prompts to high testing error, (iv) High model complexity leads to high training error and low testing error.
- (d) You are given the following data points from two different classes. [2]

Class 1 points: (2,3), (3,2), (4,2), (3,5), (5,4), (4,7)

Class 2 points: (5,2), (5,5), (5,7), (6,4), (7,9), (10,6)

What will be the nature of decision boundary between these two classes? Justify your answer.

- (e) Suppose you are given a large dataset and asked to perform the classification task. Do you think that k -NN algorithm would be an appropriate choice in this case? Explain your answer. [2]
2. (a) A person has built a classifier that yields 100% accuracy on training data. When he/she deploys his/her model on the client-side, it is found that the model is not accurate. Which thing might go wrong? Note that the model has successfully deployed, and no technical issues are found at the client-side except the model performance. [3]
- (b) Suppose X and Y are two random variables. If $V[X] = 4$ and $V[4Y] = 32$, then find the variance of $Z = 2X - 8$ and $U = Y + 2$. [3]
- (c) Suppose X be a continuous random variable with PDF [3]

$$f(x) = \begin{cases} x + \frac{1}{2}, & 0 \leq x \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

Find $E[x^n]$, where $n \in \mathbb{N}$.

- (d) Is accuracy a good measure for classification of imbalanced data? Justify your answer with one suitable example. [1]
3. (a) Consider a two-class classification task with the following samples.
- Class 1 (c_1) $\rightarrow x_1 : (0, 1), x_2 : (1, 0), x_3 : (1, 2), x_4 : (2, 1)$
- Class 2 (c_2) $\rightarrow x_5 : (3, 5), x_6 : (4, 3), x_7 : (5, 4)$
- (b) Assuming classes are distributed according to Gaussian distribution, find a class label for sample $x = [1.0, 2.2]$ into class c_1 or c_2 according to Bayesian classification rule. [6]
- (c) Suppose you are working on a binary classification problem for email spam detection. You have a dataset of 200 emails, and your classifier has the following performance:
130 emails are correctly classified as non-spam, 25 emails that are non-spam are incorrectly classified as spam, 30 emails that are spam are correctly classified as spam and 15 emails that are spam are incorrectly classified as non-spam.
Design a confusion matrix based on the above data for the email spam detection problem and calculate precision, recall and F1-score. [4]

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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTA

M.Tech. (CSE), Semester – I
Mid Term Examination, Odd Semester 2024-25

Marks: 30 (Weightage – 30%)

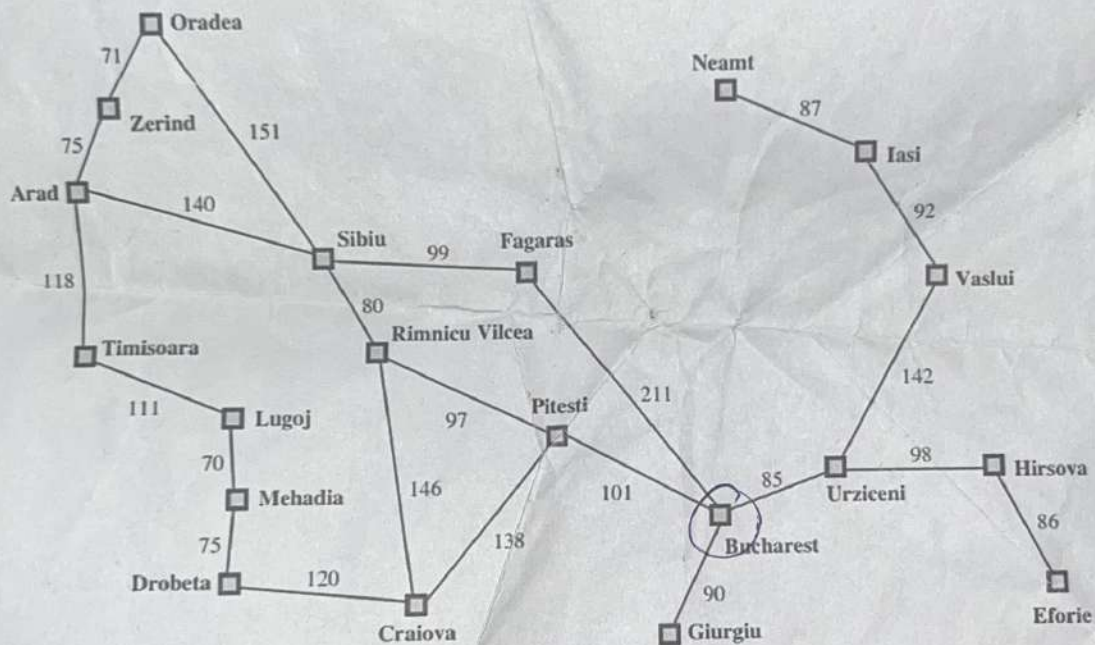
Time: 90 minutes

Date: 25-9-2024

Note: Attempt all the questions. Take suitable assumptions, if required.

Question 1: What is the working process Min-Max algorithm? Consider the case study of Tic-Tac-Toe game problem, explain the minimax algorithm step-by-step and generate the game tree. [6 marks]

Question 2: Consider the following graph and the heuristic table. Let us assume “Arad” as the starting node and “Bucharest” as the goal node. Find the shortest path using A* algorithm. Prove its optimality mathematically. [6 marks]



Graph is shown above while the heuristic distance given below.

Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

Question 3: Consider the following 8-puzzle game problem in AI. Apply greedy best first search algorithm to solve it. Show at least 2 iterations. [6 marks]

Initial State			Goal State		
1	2	3	2	8	1
8		4		4	3
7	6	5	7	6	5

Question 4: Analyze the time complexity of DFS (depth first search) and BFS (breath first search). Compare both the algorithms and discuss their applications. [6 marks]

Question 5: Explain hill climbing algorithm mathematically. [6 marks]

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M.Tech. (CSE) Semester –I
Mid Term Examination, Odd Semester 2024-25

Nature Inspired Algorithms(CST509)

Marks: 30 (Weightage – 30%)

Time: 90 minutes

Date: 26.09.2024

Q1. (a) Write the algorithm for Steepest Descent. For what kind of problems is it generally used and why? (5 marks)

(b) Perform at least three iterations of the steepest descent method for the given function:

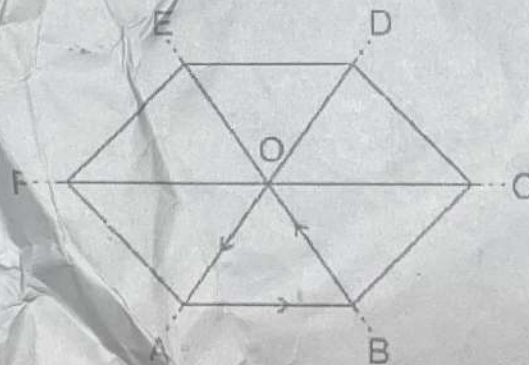
$$f(x_1, x_2) = 4x_1^2 + 6x_2^2 - 8x_1x_2 \text{ with } X_1 = (1, 1)$$

(5 marks)

Q2. (a) List 5 disadvantages of traditional optimization techniques. (5 marks)

(b) In one dimension, a random walker takes a step with equal probability to the left or right. What is the probability that he returns to the starting point after 4 steps. (2.5 marks)

(c) Consider a random walk on an infinite two-dimensional triangular lattice, a part of which is shown in the figure below.



If the probabilities of moving to any of the nearest neighbour sites are equal. What is the probability that the walker returns to the starting position at the end of exactly three steps?

(2.5 marks)

Q3. Write short notes on:

(a) Classification of optimization problems based on Objectives and Constraints (5 marks)

(b) Main features of Genetic Algorithms (GA) with flowchart (5 marks)

1. Given the initial weights as $w_0 = -2$, $w_1 = 1$ and $w_2 = 2$. The hypothesis is $h(x) = \text{sign}(w^T x) \geq 0$ is a positive class. Run one epochs of perceptron learning algorithm on the following data. (Optional If you wish you may use $\eta = 0.5$). What is the w vector after one epoch. (5)

Example	x_0	x_1	x_2	y
e_1	1	1	0	-1
e_2	1	0	1	-1
e_3	1	2	2	-1
e_4	1	2	1	-1

TABLE 1: Training set

2. Derive the closed form solution for multiple linear regression model to train a model for optimal parameter. Using the closed form method also compute the optimal parameter for multiple linear regression for the given data. (3 + 5)

Example	x_0	x_1	x_2	y
e_1	1	1	4	1
e_2	1	2	5	6
e_3	1	3	8	8
e_4	1	4	2	12

TABLE 2: Training set

3. With the help of a diagram explain overfitting? How can we overcome it. (2.5)
4. Define Bias and variance in terms of model complexity. (2.5)
5. Show that $(X^T X + \lambda I)$ is a positive semidefinite or a non singular or a full rank matrix. (2)
6. Decompose the quadratic function $f(x_1, x_2, x_3) = 5x_1^2 + 4x_2^2 - x_1 x_3 + 4x_2 x_3 + 7x_3^2$ as matrix and vector product. (2)
7. With the help of a diagram explain the supervised learning model, illustrating standard notations for: input, output, hypothesis and algorithm. (3)

Department of Computer Science and Engineering, IIT Kota
First Mid Term Examination 2024, M.Tech 1st Semester(CSE)

Subject: Advanced Algorithms

Max Marks: 30

Date and time: 14/10/2024; (10:30am - 12:00pm)
Note: Attempt THREE questions. Each question carries 10 marks.

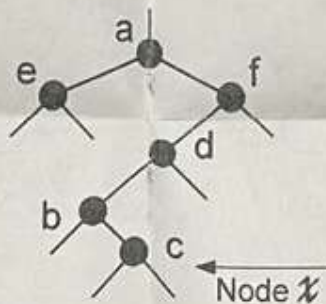
Q1 a) Answer whether **TRUE** or **FALSE** with justifications :

- The running time of Quicksort is $T(n) = \Theta(n \log n)$, when all the elements are the same. F
- Amortized analysis determine the worst-case bound on sequence of operations without using probability.
- If we had an algorithm to square a n -bit number in $O(n)$ time, we could multiply two n -bit numbers in $O(n)$ time.
- For any n -node Binary Search Tree T , the increase in potential due to insertion of a new leaf into T , excluding the cost of splaying, is at most $n \log n$.
- $f(n) = O(g(n))$ if and only if $g(n) = \Omega(f(n))$

(b) Solve the recurrence: $T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n-1) + n(n-1) & \text{if } n \geq 2 \end{cases}$

Q2. Answer the following related to Splay Tree:

- Apply the $\text{Splay}(x, S)$ operations for the following tree to make the node x as a root of the Splay tree



- Let $S(x)$ be the sub-tree of S rooted at x and $\mu(S)$ and $\mu(x) = \log(n)$ the ranks of S and node x respectively. Prove Splay Lemma's:
Splay lemma (ZIG) takes $\leq 3(\mu(S) - \mu(x)) + 1$ credits to perform a ZIG operation to maintain Splay invariant.

Q3. Construct a simple binary search tree (BST) containing:

12 5 24 20 3 30 10 55

- Construct then a Red black tree (RB tree) and
- Convert the constructed RB Tree into 2-3-4 tree
- Insert the following numbers in RB Tree 6 and 25
- Delete the number 22 from the RB Tree

- Q4. Given a chain of matrices $(A_1, A_2, \dots, A_n)$, where A_i has dimensions $p_{i-1} \times p_i$, find a way to parenthesize the product $A_1 \cdot A_2 \cdot \dots \cdot A_n$ in a way that minimizes the number of scalar multiplications.
- (a) Give the recursive solution for minimizing the cost of parenthesizing the matrix $A_i \cdot A_{i+1} \dots A_j$ and
- (b) Compute $A_1 \times A_2 \times A_3$ where: $A_1: 10 \times 45$; $A_2: 45 \times 25$; $A_3: 25 \times 50$

- Q5. An automobile factory has two assembly lines as shown in Fig-1. Each line has n stations: $S_{1,1}, \dots, S_{1,n}$ and $S_{2,1}, \dots, S_{2,n}$. Corresponding stations $S_{1,j}$ and $S_{2,j}$ perform the same function but can take different amounts of time $a_{1,j}$ and $a_{2,j}$. Entry times are: e_1 and e_2 ; exit times are: x_1 and x_2 . After going through a station, a car can either: stay on same line at no cost, or transfer to other line: cost after $S_{i,j}$ is $t_{i,j}$, $j = 1, \dots, n-1$ as mentioned in Fig-1
- (a) Give the recursive solution for the optimal solution to sub-problems in order to minimize the total time through the factory for one car
- (b) Determine the optimal solution in terms of the optimal solution to sub-problems.

